



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

January 10, 2023

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours

Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 30 min before the end of the exam you will get 3 points, if you finish earlier than 20 min you will get 2 points and if you finish earlier than 10 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Describe and compare Q-learning and SARSA.

Exercise 2 (7 marks)

What is the **VC dimension** of a hypothesis space? What can it be used for?

Exercise 3 (2 marks)

Consider the following snippet of code:

```
1 coef = []
2 mse = []
3 for alpha in [0.001, 0.01, 0.1, 0.2, 0.5]:
4     model = linear_model.Lasso(alpha=alpha)
5     model.fit(x, y)
6     mse.append(mean_squared_error(y, model.predict(x)))
7     coef.append(model.coef_)
8 i = np.argmin(mse)
9 print(i, mse[i], coef[i])
```

1. Describe the procedure provided by the snippet above. Is it correct? If not, propose a correction.
2. What is the purpose of the procedure above? Can you tell which property have the coefficients `coef` computed in the snippet above?
3. Can you list at least two other approaches that can be used for purposes similar to the one provided by the procedure above? Motivate your answers.

1. The above procedure is evaluating the regularization parameter for a LASSO model for the problem of regression. It fits models with different values of α and computes the training RSS to decide which one is the most suitable for the given dataset. Clearly this procedure would give as a result that the model with the smallest parameter $\alpha = 0.001$ is the one to choose, since we are validating our possible models on the training set. A correct procedure would have to validate on independent data (for instance, using validation or crossvalidation).
2. The purpose of the procedure is to introduce regularization into a regression model to avoid overfitting. The LASSO procedure produces a final parameter vector which is sparse. Indeed in Line 5 the resulting printed parameter would have less and less non-zero elements as the value of α (λ) increases.
3. To avoid overfitting one may apply other regularization techniques, like Ridge or model selection techniques, like the backward/forward feature selection approach. Differently from the LASSO we would have that the final solution is not necessarily sparse with Ridge, while it might be with the wrapper approaches.

Exercise 4 (2 marks)

Tell if the following statement about the Principal Component Analysis (PCA) procedure are true or false. Motivate your answers.

1. The set of the Principal Components vectors are providing an orthonormal base for the original feature space.
2. Using as features for regression/classification problems the projection of the original features into the principal components provided by the PCA reduces the phenomenon of overfitting.
3. The percentage of the variance explained by a Principal Component is inversely proportional to the value of the corresponding eigenvalue.
4. The procedure to apply PCA to a dataset is deterministic.

1. TRUE: The procedure for the PCA looks for the direction in the dataset which is providing the most variance and extracts the (first) Principal Component (PC) as the unit vector identifying that direction. Iteratively, it checks the direction, orthogonal to the previous one, with the most variance and extracts another (second) PC. The process iterates over a number of PC equal to the number of dimensions of the dataset. This produces an orthogonal base to the initial dataset.
2. FALSE: only selecting the first K PC one has the chance to remove the noise from the data and avoid overfitting. If we are keeping all the PC, we would have a linear transformation of the original dataset, which is likely to behave as good as the original one for the supervised task.
3. FALSE: The variance explained by each PC is directly proportional to the corresponding eigenvalue. Indeed, its formula is $\frac{\lambda_i}{\sum_i \lambda_i}$, where λ_i is the eigenvalues corresponding to the i -th PC.
4. TRUE: The procedure includes a mean-scaling procedure and the inversion of the covariance matrix, which are both deterministic procedures.

Exercise 5 (2 marks)

Tell if the following statement about the Gaussian Processes (GP) are true or false. Motivate your answers.

1. The main difference between the Nadaraya-Watson Model and the GP is that the probabilistic model is defined over the joint input/output for the former model and only on the output for the latter model.
2. The GP model is a linear model w.r.t. the optimal parameter vector learned using a maximum likelihood estimation procedure.
3. Assuming to have the inverse of the Gram matrix, the computational complexity of providing a new prediction is quadratic in the number of samples present in the training set.
4. The prediction provided by a GP with a linear kernel, i.e., $k(x, y) = x^\top y$, corresponds to the one provided by a specific linear regression model.

1. TRUE: The Nadaraya-Watson Model imposes a prior on $p(x, y)$, where x are the input and y is the output, while GPs are only taking into account a probabilistic model on the target y . They both derive the value of the mean prediction starting from such priors.
2. FALSE: GPs are nonparametric models, and therefore it is not always possible (see answer to 5.4) to define a parameter vector summarizing the GP model. Instead, the model is given by the training dataset itself.
3. TRUE: The inversion of the matrix would have costed $O(n^3)$, where n is the number of samples present in the dataset. Instead, the formula to compute the mean prediction includes a vector/matrix multiplication and a vector/vector multiplication, which are $O(n^2)$ and $O(n)$, respectively.
4. TRUE: The final formula would be $m(x) = x^\top X C t = x^\top \theta$, where X is the matrix of the training inputs. Note that the value of θ is entirely dependent on the dataset.

Exercise 6 (2 marks)

You are an ML specialist working for an online shipper that accepts orders and delivers them relying on several trucks. You are facing the following problems:

1. Estimating the shipping time, based on the order characteristics (e.g., arrival time, weight, dimensions, pick up location, destination location).
2. Identifying which are the characteristics of the order that are irrelevant for accomplishing the task described in the previous point.
3. Deciding the price to charge the client in order to maximize the long-term profit, accounting also for the customer fidelization phenomenon.

Model these three problems as ML problems, specifying the data (input, output, state, action, reward), the problem class, and a specific method to solve them.

1. This is a SUPERVISED LEARNING problem, specifically, MULTIVARIATE REGRESSION. The input variables are the order characteristics (e.g., arrival time, weight, dimensions, pick up location, destination location) and the output variable is the shipping time (continuous variable). A possible method for solving it is linear regression.
2. This is a FEATURE SELECTION problem. The input variables are the order characteristics (e.g., arrival time, weight, dimensions, pick up location, destination location). The output variable might be missing in the case of unsupervised feature selection or corresponding to the shipping time for supervised feature selection. A possible method for solving it is a wrapper approach that uses linear regression as inner algorithm and forward feature selection.
3. This is a REINFORCEMENT LEARNING problem. The state is represented by the historical information about customers and orders (from which deriving information about the fidelization) as well as the characteristics of the current order. The action is the price to be assigned to the current order. The reward is the profit (revenue - costs) related to the current order. It would have been inappropriate to consider the problem a MULTI-ARMED BANDIT because of the fidelization phenomenon that is such that our actions (prices) determine a modification of the environment state. A possible method for solving it is Q-learning.

Exercise 7 (4 marks)

Consider the linear two-class SVM classifier defined by the parameters $w = [5, 3]$ and $b = -3$ and trained on a **linearly separable** dataset. Answer the following questions, providing adequate motivation.

1. How is the point $x_1 = [2, -3]$ classified according to the trained SVM?
2. Suppose that the true label of x_1 is $t_1 = +1$. Is it possible that x_1 was contained in the training set? What about the case $t_1 = -1$?
3. Consider the set of points of the form $x = [a, 2]$. For which values of a the point x lies on the margin?

1. Let us compute the expression $w^T x_1 + b = 5 \cdot 2 + 3 \cdot (-3) - 3 = -2 < 0$. Thus, the point is classified as a negative instance.
2. If $t_1 = 1$, x_1 cannot be contained in the training set, since, due to the previous calculation, it would have been misclassified, violating the linear separability assumption. If, instead, $t_1 = -1$, then x_1 is correctly classified and, consequently, could have been contained in the training set.
3. A point x lies on the margin if $|w^T x + b| = 1$. Thus, we have:

$$1 = |[5, 3]^T [a, 2] - 3| = |5a + 6 - 3| \implies a \in \left\{ -\frac{2}{5}, -\frac{4}{5} \right\}.$$

Exercise 8 (4 marks)

Assume to have a stochastic Multi-Armed Bandit (MAB) with 3 arms with average reward:

$$R(a_1) = 0.1,$$

$$R(a_2) = 0.6,$$

$$R(a_3) = 0.3,$$

and each distribution $\mathcal{R}(a_i)$ is a Bernoulli.

1. Write the asymptotic minimum expected pseudo-regret we might have on average over $T = e^{10}$ time steps.
2. If we apply the UCB1 algorithm, which is the upper bound on the expected pseudo-regret? Is it larger or smaller than the previous one?
3. What can we tell about the minimum regret we might have on the above problem? And about the minimum regret of UCB1?

Note that the KL divergence for Bernoulli variables with means p and q is:

$$KL(p, q) = p \log \left(\frac{p}{q} \right) + (1 - p) \log \left(\frac{(1 - p)}{(1 - q)} \right)$$

and (if necessary) that $\log(\frac{0.1}{0.6}) = -1.8$, $\log(\frac{0.9}{0.4}) = 0.8$, $\log(\frac{0.3}{0.6}) = -0.7$, $\log(\frac{0.7}{0.4}) = 0.6$, $\log(\frac{0.1}{0.3}) = -1.1$, and $\log(\frac{0.9}{0.7}) = 0.25$.

1. The lower bound for a stochastic MAB problem is (for T sufficiently large):

$$L_T \geq \log T \frac{\Delta_i}{\sum_{i, i \neq *} KL(\mu_i, \mu^*)},$$

where μ^* is the expected reward of the optimal arm, μ_i is the expected reward of the i -th arm, and $\Delta_i = \mu^* - \mu_i$. In our case we have:

$$\begin{aligned} L_T &> 10 \left(\frac{0.6 - 0.1}{KL(0.1, 0.6)} + \frac{0.6 - 0.3}{KL(0.3, 0.6)} \right) \\ &= 8 \cdot 10 \left(\frac{0.5}{(0.1 \cdot (-1.8) + 0.9 \cdot 0.8)} + \frac{0.3}{(0.3 \cdot (-0.7) + 0.7 \cdot 0.6)} \right) \approx 23.5 \end{aligned}$$

2. The formula for the upper bound of the UCB1 algorithm is:

$$L_T \leq 8 \log T \sum_{i \neq *} \frac{1}{\Delta_i} + 1 + \frac{\pi^2}{3} \left(\sum_{i \neq *} \Delta_i \right),$$

which in our setting becomes:

$$\begin{aligned} L_t &\leq 10 \left(\frac{1}{0.6 - 0.1} + \frac{1}{0.6 - 0.3} \right) + \left(1 + \frac{\pi^2}{3} \right) (0.6 - 0.1 + 0.6 - 0.3) \\ &= 80 \left(2 + \frac{10}{3} \right) + \left(1 + \frac{\pi^2}{3} \right) \cdot 0.8 \approx 430. \end{aligned}$$

3. All the above theoretical results are only holding in expected value. Both bounds can be violated on specific runs of the UCB1 algorithm.

